

OR Deliverable 2: Fashion Parsing

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Introduction

This report documents the development of a fashion parsing system using the Fashionpedia dataset, a large-scale image repository designed for fine-grained clothing segmentation. Image segmentation is the process of partitioning an image into multiple sets of pixels to simplify its visual representation. Specifically, this project focuses on semantic segmentation, where every pixel is assigned a categorical label like "top," "hair," or "skin" to identify the class of every object in the scene and instance segmentation, which identifies every individual object as a separate entity.

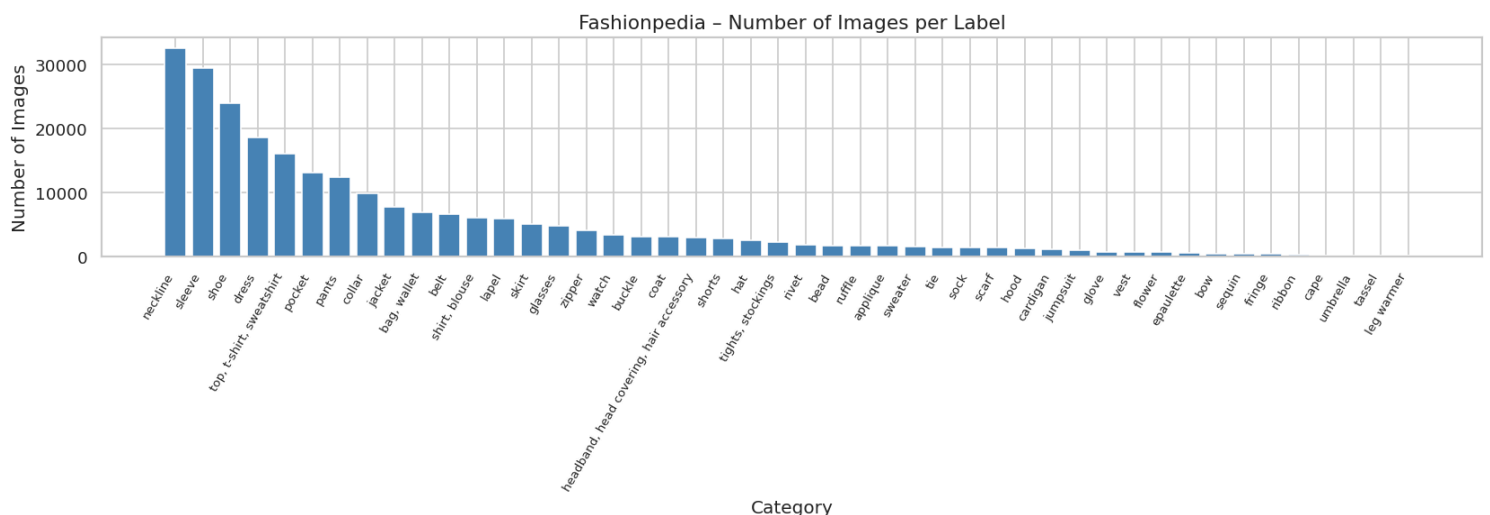
The task follows a structured experimental flow, beginning with an exploratory analysis of label distributions by calculating image counts and pixel-to-foreground ratios. Following this data analysis, three different architectures from the MMSegmentation library are selected and finetuned to evaluate the impact of hyperparameters, data augmentation, and input resolution on segmentation accuracy. Also, we assess the impact of discarding overrepresented labels on model performance. Then we present a comparative study against a YOLO-based instance segmentation model. The results are evaluated using both qualitative visualizations and quantitative metrics, specifically accuracy and mDice. The report concludes with a theoretical discussion of two potential strategies to improve results, backed by state of the art research

Dataset Analysis

In this section we evaluate the statistical properties of the Fashionpedia dataset. The analysis focuses on the quantitative distribution of labels and their spatial characteristics. Fashionpedia is a large dataset designed for the task of instance segmentation and fine-grained attribute recognition. Unlike previous fashion datasets that focus only on high-level categories (e.g., "dress" or "pants"), Fashionpedia uses a complex, expert-built ontology. This ontology consists of 46 object categories, divided into 27 main apparel items and 19 apparel parts alongside 294 fine-grained attributes. The dataset contains approximately 48,000 images from diverse real-world contexts, including street-style photography, celebrity events, and online retail. Each image is annotated with per-mask segmentation.

Label Distribution Analysis

The dataset has a significant class imbalance as seen in figure 1 by a long-tail distribution. For example neckline (n=32,590) and sleeve (n=29,397) are the most occurring labels. Main categories like top, t-shirt, sweatshirt (n=16,102) show good representation, providing enough samples for generalized feature learning.



Specialized garments and accessories, like cardigan ($n=1,101$) and buckle ($n=3,099$), are in the tail of the distribution.

Spatial Characteristics and Pixel Ratios

The physical area an object takes up in an image varies drastically across classes. This is analyzed through two metrics: the ratio relative to the full image and the ratio relative to the detected foreground.

Categories such as sweater and shirt, blouse represent the largest spatial footprints. A sweater occupies a mean of 15.1% of the total image and 42.8% of the foreground. The high $\mu_{foreground}$ values suggest that when these items are present, they dominate the visual field. However, the variance ($\sigma^2 \approx 0.073$ for t-shirts) indicates significant diversity in how these items are captured. This variability is likely because of differences in camera distance and garment styling (e.g., oversized vs. slim), requiring the model to be robust to intra-class scale transformations.

The most frequent labels often have the smallest spatial footprints. Despite being the most frequent label, a neckline takes up only 0.27% of the full image. Buckle is even smaller, at 0.08% of the image pixels.

Semantic Segmentation

Model and Training Setup

For the semantic segmentation part of the project, we evaluated three architectures on the Fashionpedia dataset: DeepLabV3+ with a ResNet-50 backbone, SegFormer-B0, and UPerNet with a Swin-Tiny backbone. We chose these models to cover different design families, from a strong CNN-based baseline to more recent transformer-based approaches. All models were trained to assign a semantic class to each pixel in the image, which is good for fashion parsing, considering there's a lot of pixel coverage.

Before training, we loaded the images together with their semantic masks through a shared preprocessing pipeline. An example of the data used for training is shown below. This figure helps show the kind of supervision the models receive: each image is paired with a dense categorical mask in which every visible region is assigned a garment label. It also makes one of the main challenges of the task very clear, mainly the large variation in scale, garment overlap, and small accessory regions.

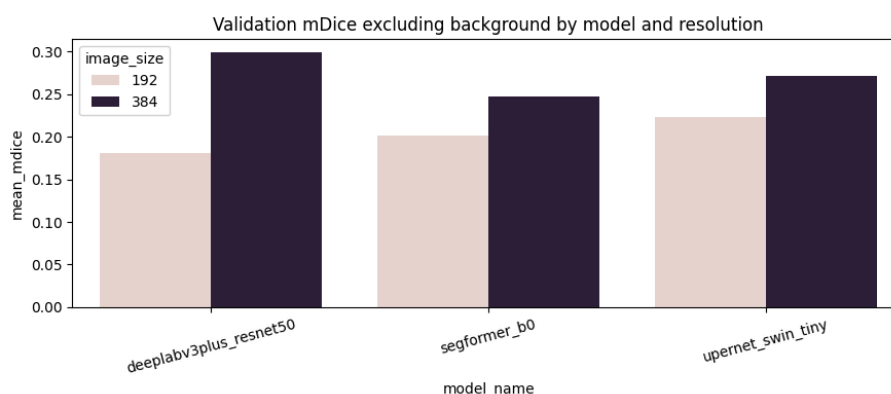


To study the effect of architectural and training choices, we explored a grid of hyperparameters including two image resolutions (192 px and 384 px), two learning rates ($1e-3$ and $3e-4$), two batch sizes, and two augmentation settings (weak and strong). We ranked the models primarily by validation mean Dice, while pixel accuracy and mean IoU were used as supporting metrics. All of these metrics were calculated excluding the background.

Quantitative Results

The semantic segmentation experiments gave competitive results across all three model families. Among all tested configurations, the best result was achieved by UPerNet with Swin-Tiny at 384 px, with `learning_rate = 3e-4`, `batch_size = 16`, and weak augmentation. This model reached a validation mDice excluding background of 0.364, a pixel accuracy excluding background of 0.731, and a mIoU excluding background of 0.261. These results made it the strongest semantic model in the full experiment table.

The overall comparison between the three architectures is illustrated in figure below. We use this figure to show that UPerNet gave the strongest overall result, while DeepLabV3+ remained competitive and SegFormer-B0 produced the weakest average performance, although still above the assignment threshold. The figure also helps show that the differences were not only from the architecture but were also influenced by input resolution and the training configuration.



Effect of Resolution and Augmentation

The higher resolution setting, 384 px, consistently outperformed 192 px in the strongest runs. This suggests that the additional spatial detail is beneficial for Fashionpedia, especially for classes with thin boundaries, fine garment contours, or small accessory regions. Since fashion parsing often depends on accurately separating visually adjacent items such as jackets, shirts, collars, or shoes, this result fits very naturally with the structure of the dataset.

The effect of augmentation was more model-dependent. In general, weak augmentation turned out to be the safer choice for the top-performing runs, particularly for SegFormer and UPerNet. While DeepLabV3+ showed somewhat more robustness under stronger augmentation, its best run still remained below the strongest UPerNet configuration. This suggests that moderate augmentation was enough to improve generalization without introducing too much appearance distortion for this task.

This trend can be summarized in the aforementioned figure and the effect of augmentation can be seen in the table below.

Model Name	Augmentation Type	Mean mDice
deeplabv3plus_resnet50	strong	0.252894
deeplabv3plus_resnet50	weak	0.218725
segformer_b0	strong	0.205289
segformer_b0	weak	0.248301
upernet_swin_tiny	strong	0.238129
upernet_swin_tiny	weak	0.252437

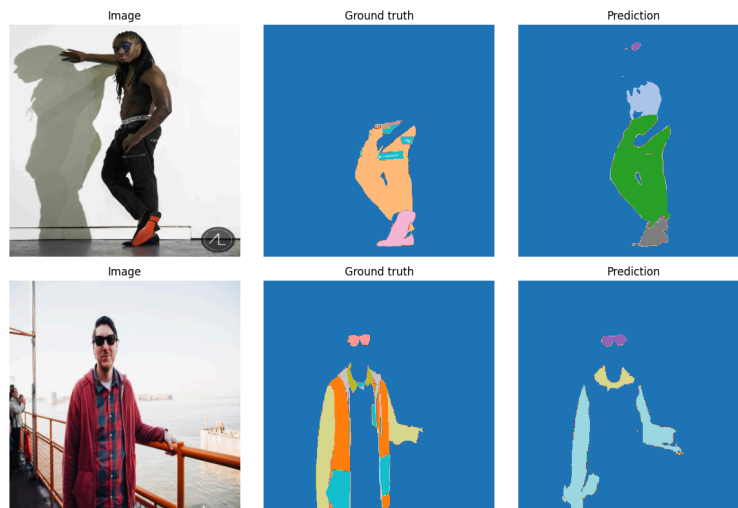
Per-Class Performance Analysis

Even though the global validation metrics are useful, they don't fully capture the difficulty across the different fashion labels. Per-class analysis showed us that large and visually distinctive garments tend to achieve higher Dice scores, while small, rare, or visually ambiguous classes remained more challenging. Categories such as pants, dress, or umbrella were segmented relatively well by the best semantic model, however, smaller accessory-like or fine detail classes were much less reliable. This class-dependence pattern is coherent with the dataset analysis presented earlier in the report. Labels with many examples and larger pixel footprints naturally provide a stronger training signal, while classes that occupy only a tiny fraction of the image are much harder to learn. In practice, this means that a reasonable overall mDice can still hide substantial weaknesses on the long tail of the label distribution.

Qualitative Results

The predictions of the best full-label UPerNet model are shown below. This figure shows how the model behaves on real validation examples. In general, the qualitative predictions confirm the quantitative trends: large garments are often segmented with coherent shapes and good boundary alignment, while more difficult regions tend to be partially missed, merged with nearby labels, or assigned unstable boundaries.

These examples are important because they show both the strengths and the limitations of the model more clearly than average metrics alone. Even in cases where the predicted segmentation is not perfect, the model usually captures the dominant garment regions correctly, which supports the conclusion that UPerNet is an effective dense parser for the main Fashionpedia categories. At the same time, the visualizations reveal where fine-grained distinctions remain difficult, especially when labels are small, visually subtle, or partially occluded.



Visualization of UPerNet Predictions vs. Ground truth.

Reduced-Label Finetuning

After identifying the best semantic segmentation model, we studied the effect of discarding large and overrepresented labels. The reduced-label experiment kept the same architecture and core hyperparameters as the best full-label model, namely UPerNet with Swin-Tiny at 384 px, with `learning_rate = 3e-4`, `batch_size = 16`, and weak augmentation. We adapted the output head to the reduced label space and reused the compatible weights from the best checkpoint.

The full-label baseline achieved a validation mDice excluding background of 0.364, while the reduced-label finetune reached 0.326. The reduced-label run therefore did not surpass the best full-label model overall. Even though the reduced-label finetune did not improve the global semantic segmentation result, its effect at the class level was not totally negative. Some retained categories improved after finetuning on the reduced label set, while others became worse. This suggests that removing dominant labels can free some modelling capacity for underrepresented classes, but the benefit is not universal.

One likely reason is that removing dominant classes reduces the amount of useful training signal, since those classes contribute heavily to the gradients that help the backbone learn strong visual features. Additionally, reusing weights from a checkpoint trained on a different label distribution creates a mismatch: the backbone was optimized for a broader set of classes and may not fully adapt to the reduced label space within the available training steps. This is related to catastrophic forgetting, where the model partially loses useful representations without having enough data or training time to recover them. A further factor is that the global mDice averages performance across all retained classes, so any categories that became worse after finetuning can pull the overall score down significantly, even if other classes improved.

Overall, the reduced-label finetuning experiment shows that removing overrepresented classes can help some remaining labels, but it does not produce a better global model than the original full-label UPerNet configuration.

Semantic Segmentation Summary

Overall, the semantic segmentation experiments show that all three architectures are viable for Fashionpedia, but UPerNet with Swin-Tiny at 384 px gave the strongest overall performance. The experiments also show

that resolution mattered more than strong augmentation, and that global performance remained heavily influenced by label imbalance and object size. The qualitative predictions confirm that the best semantic model is effective at garment parsing, especially for the dominant classes, but still struggles with rare and fine-detail labels. The reduced-label finetuning study reinforces this conclusion. While it improves a subset of retained labels, it does not outperform the best full-label model as a complete parsing system. For that reason, in the semantic segmentation component of this deliverable, the best full-label UPerNet configuration should be considered the main result, with the reduced-label finetune serving as a useful secondary analysis of class imbalance effects.

YOLO-Based Instance Segmentation

Model and Training Setup

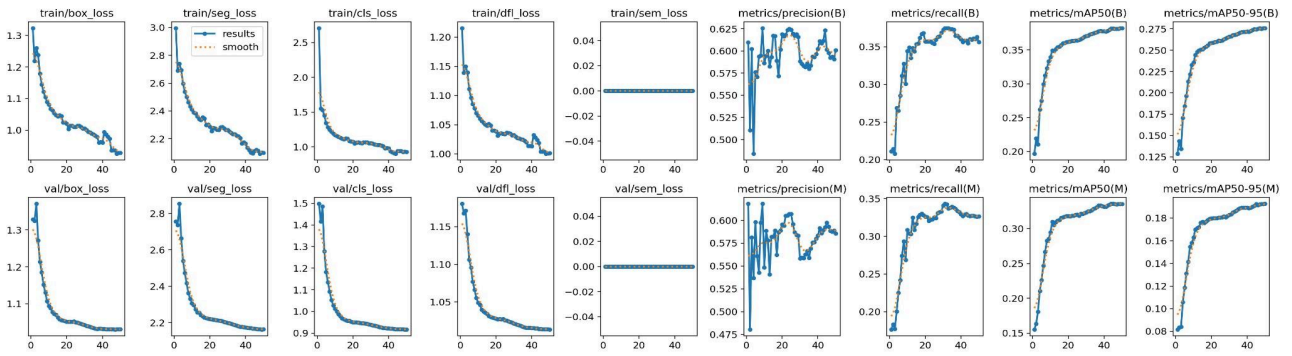
To complement and compare with the semantic segmentation experiments, we trained a YOLOv8n-seg model (nano variant, pre-trained on COCO) for instance segmentation on the Fashionpedia dataset. Unlike semantic segmentation (which assigns one class per pixel) instance segmentation detects individual objects with bounding boxes and per-instance masks, allowing it to distinguish overlapping items, for example, two separate sleeves or a jacket over a shirt.

The model was finetuned for 50 epochs at 384×384 px with batch size 47. We used AdamW (YOLO’s default auto optimizer) with an initial learning rate of 0.01 cosine annealed to a final factor of 0.01, and 3 warmup epochs. Augmentation included mosaic (disabled for the last 10 epochs), horizontal flip, HSV jitter, scale jitter ($\pm 50\%$), and random erasing ($p=0.4$). The training used mixed precision and deterministic seeding (seed=42).

Results and Training Dynamics

To enable a direct comparison with the semantic segmentation models, we converted YOLO’s instance-level predictions into dense pixel-level label maps and computed mDice, pixel accuracy, and mIoU (all excluding background). The model achieves an mDice of 0.299, pixel accuracy of 0.673, and mIoU of 0.204. In terms of instance-level metrics, mask mAP50 is 0.333 and mAP50-95 is 0.186. Mask precision sits at 58.6% while recall is considerably lower at 32.7%, meaning predictions are usually correct but many instances are missed, particularly for rare and small categories.

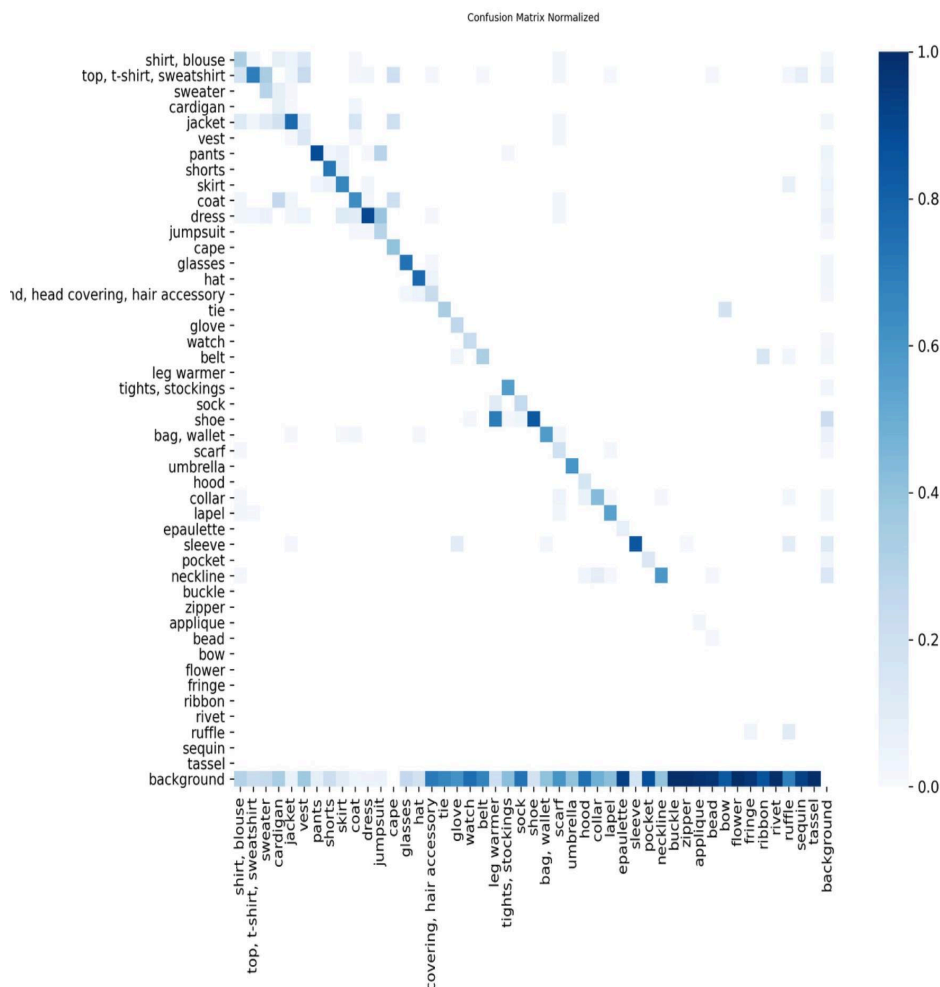
The training curves below show all losses decreasing steadily with no overfitting. However, improvement plateaus after epoch 30: mask mAP50 rises sharply from 0.155 (epoch 1) to 0.295 (epoch 10), but then only moves from 0.323 (epoch 30) to 0.335 (epoch 50), suggesting that further gains would require architectural or data-level changes rather than more epochs.



Training/validation losses and metrics over 50 epochs.

Per-Class and Qualitative Analysis

The normalised confusion matrix reveals a clear split. Large, well-represented categories (dress, pants, jacket, shoe, sleeve) show strong diagonal entries. In contrast, decorative part-level classes (buckle, zipper, applique, bead, bow, flower, fringe, ribbon, rivet, ruffle, sequin, tassel) are almost entirely classified as background, consistent with their tiny spatial footprint (e.g., buckle at 0.08% of the image). Some inter-class confusion also appears between visually similar garments like jacket/coat.



Normalised confusion matrix for YOLOv8n-seg on the validation set.

Qualitatively, the model produces clean masks with high confidence (0.7–0.9) for prominent garments like dresses, pants, and tops. The main failure modes are missed small accessories, duplicate or fragmented detections in cluttered scenes, and occasional low-confidence misclassifications between similar categories (e.g., dress vs skirt).



YOLOv8n-seg predictions on a validation batch.

Comparison with Semantic Segmentation Models

Since we converted YOLO’s instance masks into a dense label map, we can compare all models directly using mDice and pixel accuracy (both excluding background).

All four models exceed the 20% mDice threshold. UPerNet leads with an mDice of 0.364 and pixel accuracy of 0.731, while YOLO achieves 0.299 and 0.673 respectively. The gap is expected: semantic segmentation models are specifically designed for dense per-pixel prediction, whereas YOLO’s masks are generated as a secondary output to object detection. YOLO’s lower recall (32.7%) means many instances go undetected, which directly hurts mDice since undetected regions default to background.

Both paradigms share the same per-class pattern — strong performance on large garments (UPerNet Dice >0.80 for umbrella, pants, dress; YOLO confidence 0.7–0.9 on the same classes) and poor results on tail classes. Where YOLO offers an advantage is in handling overlapping garments: a jacket over a shirt produces two separate masks, which semantic segmentation cannot represent since each pixel gets exactly one label. However, for fashion parsing where complete pixel coverage matters, semantic segmentation with UPerNet remains the stronger approach overall.

Future Improvements

The first idea is to replace the standard per-pixel cross-entropy loss with a class-balanced or decoupled training strategy drawn from recent long-tail segmentation research. Methods like Seesaw Loss (Wang et al., CVPR 2021) and Balanced Group Softmax adjust the contribution of frequent versus rare classes during training, which helps prevent dominant labels such as "top/shirt" or "pants" from overwhelming the gradient

signal. More recent work in the OpenSeg line (Ghiasi et al., ECCV 2022) shows that aligning pixel embeddings with text embeddings from a pretrained language model can improve generalization to underrepresented categories without collecting extra annotations. Applied to Fashionpedia, either approach could meaningfully improve mDice on tail classes like "glove," "cape," or "headband" that currently suffer from limited training samples.

A second idea is to adopt a two-stage coarse-to-fine prediction scheme. Instead of producing all labels in a single forward pass, the model would first segment the image into broad super-categories (upper body, lower body, accessories, footwear) and then apply a second, specialized head to distinguish fine-grained labels within each group. This kind of hierarchical decoding has shown strong results in Mask2Former (Cheng et al., CVPR 2022), where a set of learnable queries attend to different regions of the feature map and are progressively refined through transformer decoder layers. Because Fashionpedia already defines a natural category hierarchy, the model could exploit that structure to reduce confusion between visually similar labels (for example, "jacket" vs. "coat") while keeping the architecture end-to-end trainable. Combining this with resolution-aware training at 384 px or higher could help the model capture small accessories that tend to disappear at lower resolutions.

Conclusion

This report presented a fashion parsing pipeline on the Fashionpedia dataset, covering dataset analysis, semantic segmentation with three architectures, and instance segmentation with YOLO. The dataset analysis confirmed a strong long-tail distribution and an inverse relationship between label frequency and spatial footprint, both of which shaped model performance throughout the experiments. Among the semantic segmentation models, UPerNet with Swin-Tiny at 384 px achieved the best results with an mDice of 0.364, while DeepLabV3+ and SegFormer-B0 also exceeded the 20% threshold. Higher resolution consistently helped across all architectures, particularly for small and thin garment classes. The reduced-label fine tuning experiment showed mixed results: some retained classes improved, but the overall mDice dropped, suggesting that removing dominant labels is not a direct path to better tail-class performance. The YOLO instance segmentation model achieved an mDice of 0.299 after converting its predictions to dense label maps, falling below the semantic models but offering the unique ability to handle overlapping garments. Across all experiments, the main bottleneck remains the long-tail class distribution: small, rare categories like buckle, cuff, and collar are difficult for every model regardless of paradigm. The future improvement ideas discussed, such as class-balanced losses and hierarchical coarse-to-fine decoding, directly target this limitation and represent the most promising directions for pushing performance further.